

E-Commerce Sales & Customer Analytics

Project Overview | End-to-End Data Analytics Workflow

Project Objective

The objective of this project was to analyse **e-commerce sales performance, customer behaviour, product performance, and inventory** and convert raw transactional data into meaningful business insights.

I used **Excel, MySQL and Power BI** to perform data transformation, analysis, KPI development, data modelling, dashboard creation and business recommendations.

Key Business Metrics

- **Total Revenue:** \$9.08M
 - **Total Orders:** 20,000
 - **Total Customers:** 5,000
 - **Units Sold:** 87,704
 - **Average Order Value:** \$454.23
 - **Repeat Customer Rate:** 84.86%
 - **Out-of-Stock Products:** 12
-

1. Data Preparation & Power Query Transformation

The first stage focused on preparing the raw data so that calculations, relationships and reporting would be reliable.

- **Imported and reviewed source tables** and checked columns, IDs, blanks and inconsistencies.
- **Changed data types** for dates, IDs, quantities and monetary fields.
- **Renamed columns** where required for consistent and readable analysis.
- **Handled missing/null values** according to the business meaning of each field.
- **Created custom columns** for customer retention/type and spending-related classifications.
- **Validated the transformed data** before loading it into the reporting layer.

2. Excel Analysis & MIS Dashboard

- Used **Power Query in Excel** for cleaning and transformation.
- Created analytical columns using **COUNTIF, SUMIF, IFS** and related Excel functions.
- Built logic for **Customer Segmentation, Customer Type/Retention and Customer Spending Distribution**.

- Used **Pivot Tables and Pivot Charts** to summarize revenue, orders, customers, products and categories.
- Added **Slicers** for interactive filtering by year, month, category and customer segment.
- Designed an executive-style **MIS dashboard** for KPI and trend reporting.

3. MySQL — Data Validation & Business Analysis

MySQL was used to validate the dataset, solve business questions and create analytical outputs.

- The **OrderDate** field was initially stored as text; it was converted into a proper **DATE/DATETIME** format for reliable date analysis.
- Used **JOINS** across Customers, Orders, OrderItems, Products and Categories.
- Applied **SUM, COUNT, AVG, GROUP BY, ORDER BY and LIMIT** to solve business queries.
- Used **subqueries and DENSE_RANK()** for product ranking, including Top 3 products by revenue within each category.
- Analyzed **top customers, top products, most-sold products, average product price, popular categories and never-purchased customers**.
- Identified **out-of-stock products** and other inventory-related opportunities.

4. Power BI — Data Model, DAX & Dashboard

- **Built relationships** between transactional and master tables to support consistent filtering and aggregation.
- Created **DAX measures** for Total Revenue, Total Orders, Total Customers, Average Order Value, Units Sold and Repeat Rate.
- Created **calculated columns** where row-level classification was required.
- Built visuals for **monthly revenue, monthly orders, revenue by category, top products, customer segmentation and spending distribution**.
- Added **slicers and filters** for Year, Month, City, Category and Customer Segment.
- Created three reporting views: **Executive Overview, Customer Analysis, and Product & Inventory Analysis**.

5. KPI Framework

Sales	Customer	Product / Inventory
Total Revenue	Total Customers	Units Sold
Total Orders	Repeat Customers	Top Products by Revenue
Average Order Value	Repeat Rate	Top Products by Quantity
Monthly Revenue Trend	New / Never Purchased	Average Product Price
	Customer Segments	Out-of-Stock Products

6. Key Findings

- **Strong customer retention:** 84.86% repeat customer rate indicates strong repeat-purchase behavior.
- **Revenue concentration:** Electronics, Home & Kitchen and Sports & Fitness contribute approximately 73.5% of total revenue.
- **Hero product:** Non-Stick Pan is a leading product across revenue, order count and quantity sold.
- **Customer activation opportunity:** 531 customers (10.62%) had never placed an order.
- **Inventory risk:** 12 of 150 products were out of stock and should be assessed against historical demand.
- **Sales decline:** Revenue and order volume decline noticeably after August and require root-cause investigation.

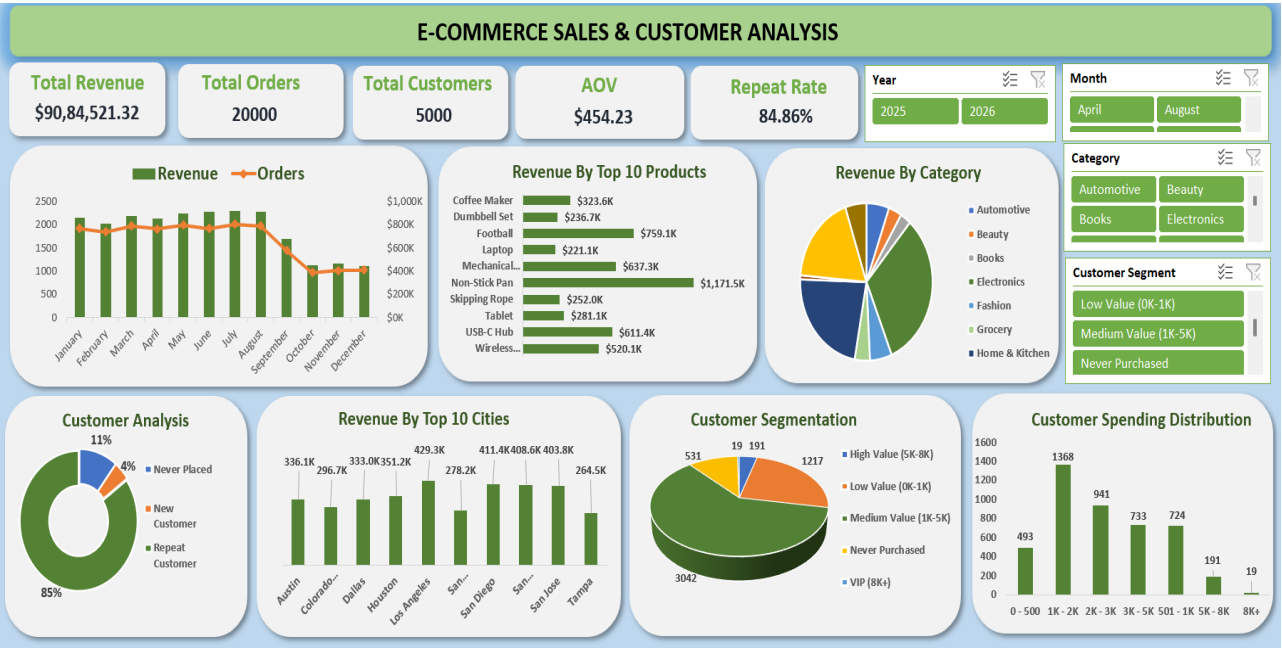
7. Business Recommendations

- **Investigate the post-August decline:** compare traffic, conversion, orders, product availability, promotions and seasonality.
- **Protect high-performing products:** prioritize inventory planning for products with high revenue and demand.
- **Increase customer value:** use cross-selling, bundles and personalized recommendations to move Medium-Value customers toward High-Value/VIP.
- **Activate never-purchased customers:** use targeted onboarding, recommendations and first-purchase campaigns.
- **Monitor category concentration:** continue investing in strong categories while improving weaker categories.
- **Improve inventory analytics:** combine historical demand, revenue and stock levels to identify potential revenue at risk.

8. Dashboard Overview & Business Intelligence

Interactive dashboards developed in Power BI and Excel to monitor sales performance, customer behaviour, product performance, and inventory.

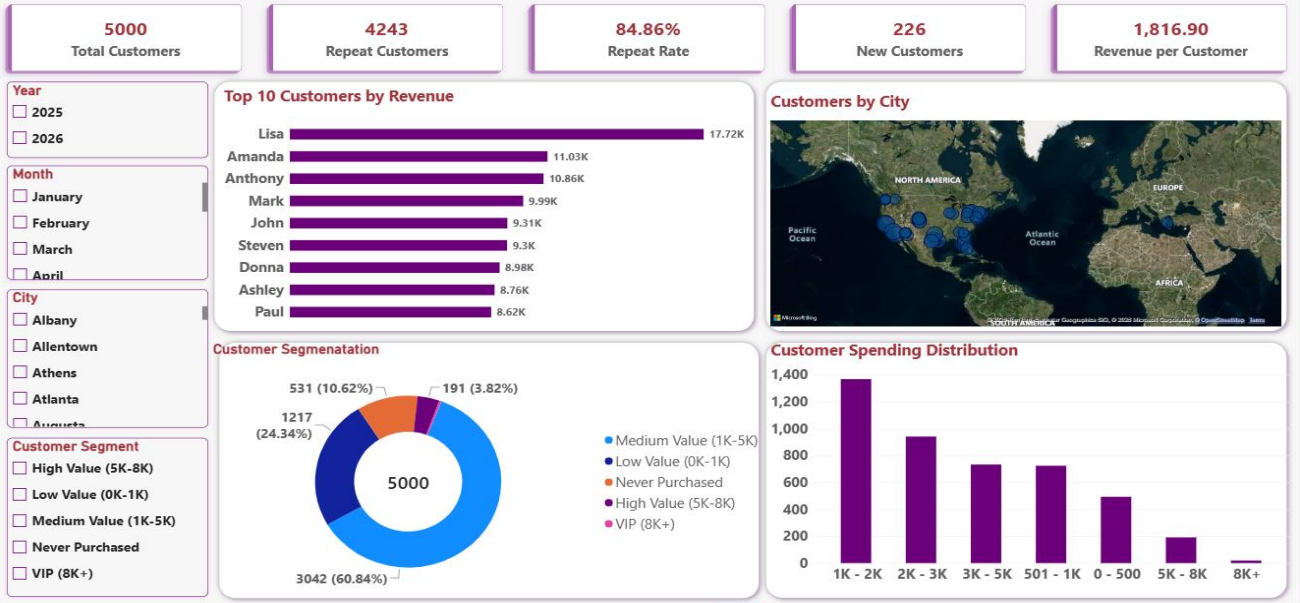
Excel Dashboard



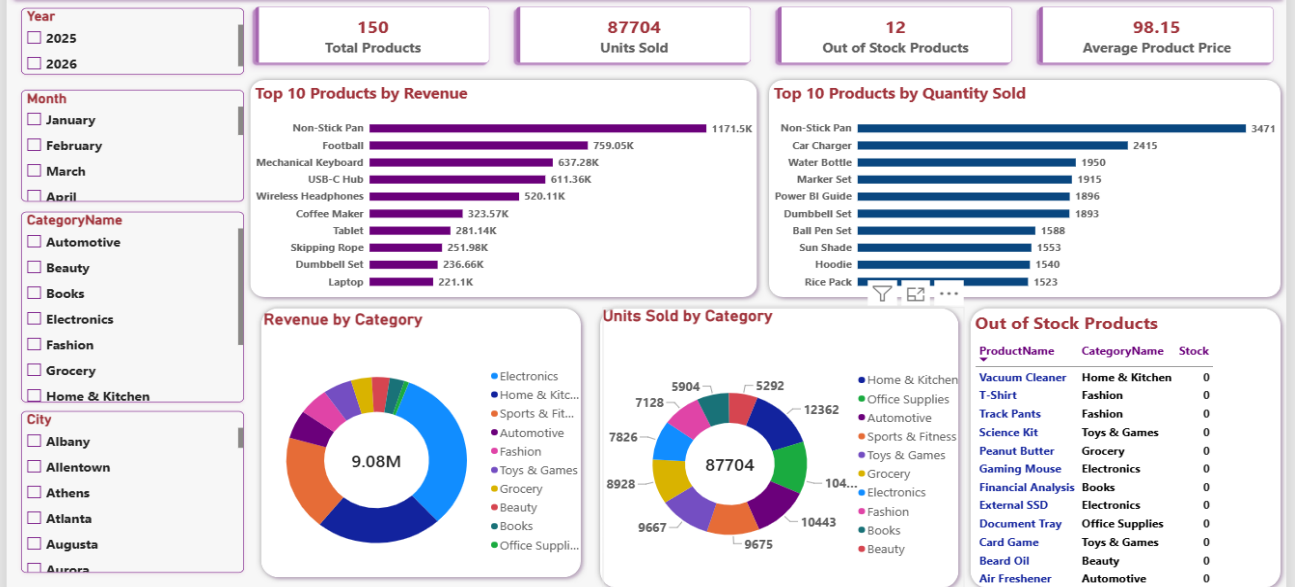
Power BI Dashboard



Page 2 - Customer Analysis



Page 3 - Product & Inventory Analysis



SQL Queries

191 # KPI 16 ... Lists products along with the Number of orders they appears in ? / Most Ordered Product

192

```
193 • SELECT p.ProductID, p.ProductName, c.CategoryName, COUNT(oi.OrderID) AS OrderCount
194 FROM OrderItems oi
195 JOIN Products p ON oi.ProductID = p.ProductID
196 JOIN Categories c ON p.CategoryID = c.CategoryID
197 GROUP BY p.ProductID, p.ProductName, c.CategoryName
198 ORDER BY OrderCount DESC;
199
```

Result Grid Filter Rows: Export: Wrap Cell Content:				
	ProductID	ProductName	CategoryName	OrderCount
▶	21	Non-Stick Pan	Home & Kitchen	2276
	142	Car Charger	Automotive	1593
	25	Water Bottle	Home & Kitchen	1305
	113	Marker Set	Office Supplies	1267
	62	Dumbbell Set	Sports & Fitness	1239
	78	Power BI Guide	Books	1231
	107	Ball Pen Set	Office Supplies	1031
	150	Sun Shade	Automotive	1026
	37	Hoodie	Fashion	1018
	134	Rice Pack	Grocery	996
	118	Paper Clips	Office Supplies	945
	27	Kitchen Scale	Home & Kitchen	917
	141	Tire Inflator	Automotive	913
	123	Breakfast Cereal	Grocery	896

60 # KPI 5 ...Retrieve the Most popular product category ?

61

```
62 • SELECT c.CategoryID, c.CategoryName, SUM(oi.Quantity) AS TotalQTYSold
63 FROM OrderItems oi
64 JOIN Categories c ON oi.CategoryID = c.CategoryID
65 GROUP BY c.CategoryID, c.CategoryName
66 ORDER BY TotalQTYSold DESC
67 LIMIT 1;
```

Result Grid Filter Rows: Export: Wrap Cell Content: Fetch rows:			
	CategoryID	CategoryName	TotalQTYSold
▶	2	Home & Kitchen	11376

```

254 # KPI .. Find the Top 3 products by Sales/Revenue in each Category ?
255
256 • SELECT t.ProductID, t.ProductName, t.CategoryName, t.Revenue
257 FROM (SELECT p.ProductID, p.ProductName, c.CategoryName, ROUND(SUM(oi.SalesAmount), 2) AS Revenue,
258 DENSE_RANK() OVER (PARTITION BY c.CategoryName ORDER BY ROUND(SUM(oi.SalesAmount), 2) DESC) AS rnk
259 FROM OrderItems oi
260 JOIN Products p ON oi.ProductID = p.ProductID
261 JOIN Categories c ON p.CategoryID = c.CategoryID
262 GROUP BY p.ProductID, p.ProductName, c.CategoryName) AS t
263 WHERE rnk <=3;
264

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	ProductID	ProductName	CategoryName	Revenue
▶	142	Car Charger	Automotive	128937.6
	147	Steering Cover	Automotive	58775.12
	137	Dash Camera	Automotive	56146.66
	49	Shampoo	Beauty	76557.72
	46	Face Wash	Beauty	44152.32
	58	Hand Cream	Beauty	30559.75
	78	Power BI Guide	Books	50489.76
	84	Leadership Book	Books	39906.9
	79	Excel Mastery	Books	36200.25
	8	Mechanical Keyboard	Electronics	589616.1
	10	USB-C Hub	Electronics	571315.8
	1	Wireless Headphones	Electronics	482327.22
	39	Backpack	Fashion	76166.3
	37	Hoodie	Fashion	72904.56

```

148 # TOP 10 Most sold products
149

```

```

150 • SELECT p.ProductID, p.ProductName, c.CategoryName, SUM(oi.Quantity) AS TotalQTYSold
151 FROM OrderItems oi
152 JOIN Products p ON oi.ProductID = p.ProductID
153 JOIN Categories c ON p.CategoryID = c.CategoryID
154 GROUP BY p.ProductID, p.ProductName, c.CategoryName
155 ORDER BY TotalQTYSold DESC
156 LIMIT 10;
157

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

	ProductID	ProductName	CategoryName	TotalQTYSold
▶	21	Non-Stick Pan	Home & Kitchen	3192
	142	Car Charger	Automotive	2220
	25	Water Bottle	Home & Kitchen	1813
	113	Marker Set	Office Supplies	1784
	78	Power BI Guide	Books	1758
	62	Dumbbell Set	Sports & Fitness	1735
	107	Ball Pen Set	Office Supplies	1456
	150	Sun Shade	Automotive	1444
	37	Hoodie	Fashion	1434
	134	Rice Pack	Grocery	1412